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Return to Play Guidelines after Cervical Spine Injuries in American Football Athletes: A Literature-based Review

Peter R. Swiatek MD[1]
Tejas Nandurkar MS[2]
Joseph Maroon MD[3]
Robert Cantu MD[4]
Henry Feuer MD[5]
Julian Bailes MD[6]
Wellington Hsu MD[7]

1. Department of Orthopaedic Surgery, Northwestern University Feinberg School of Medicine, Chicago, IL
2. Chicago Medical School at Rosalind and Franklin University of Medicine and Science, Chicago, IL
3. Professor of Neurological Surgery, University of Pittsburgh School of Medicine, Pittsburgh, PA
4. Professor of Neurological Surgery, Boston University School of Medicine, Boston, MA
5. Neurological Spine Surgery, University of Indiana Methodist Sports Medicine, Indianapolis, IN
6. Bennett-Tarkington Chair, Department of Neurosurgery, Co-Director, NorthShore Neurological Institute, Chicago, IL
7. Professor of Orthopaedic Surgery and Neurological Surgery, Northwestern University Feinberg School of Medicine, Chicago, IL

Corresponding Author:

Peter R. Swiatek, MD
676 N. Saint Clair, Suite 1350
Chicago, Illinois 60611
Ph: 312-926-4444
Fax: 312-926-4643

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MINI ABSTRACT

We review clinical and case studies related to return to play (RTP) after cervical spine injuries in elite American football athletes and offer management guidelines. The decision for RTP requires an understanding of RTP time, risks of recurrence or re-injury, and individual characteristics, including position played and pathology on imaging.

ABSTRACT

Study Design: Literature-based review

Objective: We sought to evaluate clinical and case studies related to return to play (RTP) after cervical spine injuries in elite American football athletes and to formulate guidelines to help health care practitioners manage these conditions.

Summary of Background Data: American football athletes are at unique risk of cervical spine injury and appropriate case-by-case management of cervical spine injuries is necessary for these athletes. Despite this need, no standardized guidelines exist for RTP after cervical spine injury.

Methods: Observational or case-based articles relating to RTP after cervical spine injury in American football athletes were curated from PubMed/EMBASE databases. Primary literature published before December 1, 2019 involving National Football League (NFL) or National College Athlete Association (NCAA) athletes met inclusion criteria.

Results: The data acquisition process yielded 28 studies addressing cervical spine injuries and RTP in American football athletes. Stingers/burners were the most common injury and placed athletes at higher risk of a more severe re-injury. Transient quadriplegia, cervical stenosis, cervical disc herniation (CDH), and cervical fractures have a more significant impact on the long-term health and career longevity of the American football athlete. As such, the literature offers some guidance for management of these athletes, including average time for RTP in patients treated non-operatively, thresholds involving cervical stenosis, and postoperative recommendations after spinal decompression and/or fusion surgery.

Conclusion: Elite American football athletes are at high risk for cervical spine injury due to the nature of their sport. The decision to allow these athletes to return to play should involve an understanding of the average RTP time, the potential risks of recurrence or re-injury, and individual characteristics such as position played and pathology on imaging.

Key Words: return to play, cervical spine injury, NFL, NCAA, athlete, football, cervical stenosis, stingers, burners, transient quadriplegia, cervical fracture, ACDF, spinal fusion

Level of Evidence: 3

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INTRODUCTION

The incidence, management, and outcomes after cervical spine injuries are of major interest for athletes participating in American football. From 2000 to 2010, nearly 1,000 NFL athletes suffered a cervical spine injury, accounting for 45% of all their spinal injuries in that timespan.[1] Additionally, collegiate football athletes suffered nearly 7500 cervical spine injuries from 2009 to 2014.[2] There are a variety of diagnoses within this group with each type associated with a different degree of severity, disability, and ability to participate in practices and games. Given the significance of its impact on American football player participation and performance, health care professionals need to be aware of the available literature and opinions as it pertains to diagnosis, treatment, and proper management of these injuries.

Despite the prevalence of these conditions, there do not exist any standardized consensus guidelines for return to play. Many of the pertinent treatises for treatment in this population are based upon expert opinion only. Given the variety and complexity of cervical spine injuries that can affect players, it is likely that an algorithm that incorporates general guiding principles, the best available evidence, and consensus opinion amongst experts may be most appropriate for treatment of these players. The purpose of this study is to propose a set of evidence-based guidelines for return to play after cervical spine injury in American football athletes based upon a comprehensive literature review.

METHODS

An initial search was conducted using the MEDLINE database (PubMed, U.S. National Institutes of Health, National Library of Medicine) and the EMBASE (Elsevier, Netherlands). Articles listed for publication by December 1, 2019 met the inclusion criteria. We used keywords to identify those sources aimed at discussing return to play after cervical spine injury in American football athletes. The Boolean logic query is as follows: *('cervical spine injury' OR 'brachial plexus' OR 'cervical injury' OR 'cervical injuries' OR 'cervical strain' OR 'cervical sprain' OR 'stinger' OR 'burner' OR 'cervical stenosis' OR 'cervical cord neuropraxia' OR 'transient quadriplegia' OR 'cervical disc herniation' OR 'spear tackle spine' OR 'cervical fracture' OR 'acdf' OR 'anterior cervical discectomy and fusion') AND ('NFL' OR 'National Football League' OR 'football') AND ('return to play' OR 'rtp' OR 'return to competition' OR 'return to contact' OR 'return-to-play').*

Manual searches using the above keywords were also performed to identify additional papers discussing RTP in American football athletes with cervical injuries (Figure 1). Only articles published in English were included. Two independent reviewers completed an initial screening based upon review or title and abstracts. The final set of articles for full-text review were determined based upon the following set of inclusion: (1) discusses American football athletes participating at the National Football League (NFL) or National Collegiate Athletic Association (NCAA) levels, (2) includes at least one type of cervical spine injury or condition, (3) provides a detailed discussion of RTP after cervical injury. Exclusion criteria included qualitative literature reviews and purely editorial-based or opinion articles. Publications with minimal or no discussion of RTP were excluded. Additionally, studies focusing on sports other that professional or collegiate football, such as rugby or high school sports, were excluded.

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Patient and Public Involvement

Individual patients were not directly involved in the completion of this work; instead, this work represents a review of previously published data. Research questions were formulated based upon the need for more appropriate and evidence-based guidelines for treatment of cervical spine injuries in American football athletes.

RESULTS

A total of 28 publications met the inclusion criteria for this study (Supplementary Table 1, http://links.lww.com/BRS/B708). The pertinent injuries discussed included strains/sprains (n=4 studies), stingers/burners (n=7 studies), stenosis (n=19 studies), disc herniation (n=14 studies), and fractures (n=6 studies).

For strains/sprains (n=4 studies), these injuries comprised nearly 37% of all cervical spine injuries in NFL athletes[1] from 2000 to 2010 and 55% of collegiate football players from 2009 to 2014.[2] Strains/sprains appear to be self-limited and have been associated with minimal time away from play. For NFL athletes, strains and sprains have been associated with 6 and 10 days mean loss of playing time, respectively.[1] Similarly, more than 90% of collegiate football athletes suffering strains/sprains lost fewer than 7 days of play.[2]

Stingers/burners (n=7 studies) were the most common cause of cervical spine injury in elite American football athletes, accounting for 46%[1] and 65%[2] of all cervical injuries in NFL and collegiate athletes, respectively. NFL athletes with stingers/burners with persistent symptoms experienced an average of 15 days of playing time missed[1] and 97% of collegiate athletes with stingers/burners missed fewer than 21 days of playing time.[2]

Transient quadriplegia (n=19 studies) occurred in 7.3 per 10,000 professional football players and is closely associated with the presence of congenital cervical stenosis, which puts the player at risk with extreme flexion and extension of the cervical spine [3-10]. Historically, Torg et al suggested RTP recommendations for NFL athletes based upon 0.8 vertebral canal-to-body ratio on plain radiographs and presence of symptoms. However, in a CT-based evaluation of 80 NFL athletes, Herzog et al[11] demonstrated that nearly 50% of asymptomatic players had abnormal canal-to-body ratios, despite having average spinal canal measurements on advanced imaging. These data challenge the specificity for the usage of this test in this patient population. More recent studies have identified the sagittal MRI as the most appropriate screening tool to diagnose cervical stenosis.[12,13] For example, the term "congenital cervical stenosis" has recently been specifically defined as a patient less than 50 years of age with mid-sagittal canal diameters of < 10 mm at multiple sub-axial cervical levels measured at the pedicle on MRI T2 imaging.[14] Furthermore, Aebli et al recently reported that a sagittal canal diameter of less than 8 mm may increase the risk of spinal cord injury with minor trauma to the neck.[15] MRI evaluation can also lead to the qualitative diagnosis of stenosis, or "functional cervical spinal stenosis," that grades severity by the loss of protective cerebrospinal fluid (CSF) around the spinal cord.[6]

Cervical disc herniations (CDH) (n=14 studies), from 2000 to 2010, accounted for 5.8% of cervical spine injuries in NFL athletes resulting in an average of 85 days of lost practice and game activity.[16] Proportionately fewer collegiate football players, approximately

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2%, suffered CDH from 2009 to 2014 and nearly 30% of these athletes lost more than 3 weeks of play.[2] From a position perspective, this injury has been most commonly reported in both linebackers and defensive backs, accounting for 18% and 16% of all CDH cases in NFL athletes, respectively. Furthermore, performance outcomes after surgical treatment may be worse for players of these positions than others.[17] In general, after a single-level surgery for CDH, postoperative outcomes are favorable, with 72% successfully returning to play for an average of 29 games over a 2.8-years period, while only 46% of NFL players treated non-operatively successfully returned to play for an average of 15 games over a 1.5 years period.[18,19] Maroon et al[20] demonstrated in a smaller case series that 80% were able to return to play approximately nine months after ACDF.[21]

While some experts have suggested that NFL and other American football players suffering CDH in the upper cervical spine (C2-4) should not return to collision sports after ACDF due to the catastrophic risk of adjacent segment disease in the upper cervical region,[22] Mai et al found no significant difference in RTP for NFL athletes treated surgical for upper (C2-4) versus lower (C4-T1) CDH.[17] Because NFL athletes are more prone to developing upper cervical (C3-4) pathology compared to the general population,[1,16,23,24] the decision-making process for these types of conditions are pertinent. Rates of adjacent segment disease in post-ACDF NFL athletes have been reported to be comparable to that of the general population.[17,25] Finally, American football athletes who undergo surgical management of CDH with foraminotomy have a higher return-to-play rate and shorter recovery time, but are at higher risk of requiring re-operation.[17,26]

A total of 18 cervical fractures (n=6 studies) were reported in the NFL between 2000-2010[1]. According to Mall et al, cervical fractures were associated with a 120-day recovery period before RTP. Specific treatment of these injuries depends upon fracture pattern and stability,[27] but with regard to collegiate football, Chung reported a total of 114 cervical spine fractures from 2009 to 2014 that were all treated non-operatively.[2]

DISCUSSION

The diagnosis, management, and decision-making process surrounding cervical spinal injuries in American football athletes can be complex and must incorporate many different considerations – both short- and long-term. Even though players have been treated for these conditions for over a century, only recently has some evidence-based literature emerged to help guide health care practitioners with this challenging clinical problem. We believe that large-scale retrospective cohort, case series, and even case report studies can be incorporated with expert opinion to set forth guidelines to practitioners and players who are afflicted by cervical spinal injuries.

Cervical Strains and Sprains

Many of the cervical spine injuries in NFL athletes involve the soft tissue supporting the cervical spine and may be classified as strains (muscular), or strains (ligamentous). Patients typically present with symptoms of neck pain with pain limiting range of motion and absence of distal numbness, tingling, or paresthesias. For these injuries, despite absence of a direct neurologic insult, medical evaluation is necessary to rule out cervical instability from

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ligamentous disruption or occult fractures. For example, although not necessarily a contraindication to RTP, discovery of a bipartite atlas has been reported on standard workup of a collegiate football player presenting with a sprain/strain.[28] Guidelines for return to play after a cervical strain/sprain include pain-free full range-of-motion, normal strength, and the dearth of subluxation or abnormal curvature on cervical radiographs.[9,29] Torg et al recommends that any NFL athletes who fail one of the aforementioned criteria should be protected in a cervical collar for 2 weeks and then evaluated with either flexion/extension radiographs or MRI to evaluate for intervertebral disc pathology if clinically warranted.[9]

Stingers and Burners

Stingers or burners are defined as a type of brachial plexopathy manifesting as transient sensory and/or motor deficits in the ipsilateral upper extremity resulting from compression- or traction-type trauma.[30,31] Given the typical mechanisms of action, the upper and middle trunks arising from C5-C7 are more likely to be injured than C8-T1, which may be injured with forceful abduction and external rotation of the arm. [30] For athletes with recurrent injuries, cervical foraminal stenosis is commonly an associated pathology, as the dorsal root ganglion takes up the majority of space in the neural foramen and may be more susceptible to injury during a traumatic compression-type event.[32] In our anecdotal experience, the incidence of recurrence is also increased when players are returned to full contact before complete resolution of symptoms.

For these types of injuries, initial evaluation must rule out a more significant structural or spinal cord injury. For athletes with residual symptoms after injury such as limited ROM or neck pain, RTP should be postponed. After a first stinger with significant physical findings, evaluation should include cervical spine radiographs and MRI evaluation. CT and/or EMG (for chronic symptoms) can be considered in case-by-case instances. [29] In a modified Delphi consensus study that included 48 spine surgeons, of which 12 were either NCAA or NFL team physicians, Schroeder et al found that 84.5% of surgeons would allow RTP if symptoms lasted less than 5 minutes and would require additional evaluation for an episode lasting longer.[33] With complete resolution of symptoms, guidelines for RTP after injury are the same as for a strain/sprain.

Transient Quadriplegia and Cervical Stenosis

Transient quadriplegia from a cervical cord concussion is defined as a temporary loss of motor or sensory function in the arms and legs that can last from seconds to days following excessive compression, flexion, or extension injury of the cervical spine. [3,34-37] The presence of pre-existing cervical stenosis greatly reduces functional reserve, which increases the incidence of these symptoms with hyperextension or axial compression forces.[38] The severity of the episode also deserves consideration in the individualized return-to-play algorithm. The more severe and longer duration of symptoms, the greater ramifications and risk of neurologic sequelae with a repeat injury.

There is a significant amount of controversy as it pertains to the diagnosis of cervical stenosis and the proper and safe return-to-play criteria[39] for athletes of all sports. Based on the available literature to date, we believe that practitioners should rely upon four categories

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of information for the management of these conditions: clinical symptoms, physical examination, sport and position played, and imaging characteristics.[40] In the setting of a player without any residual symptoms and normal strength and neurologic examination, individualized recommendations should rely upon the type of demands of the player's respective sport as well as what the advanced imaging (e.g, MRI) studies demonstrate. The categorization of collision (i.e. American football, rugby, hockey, etc.) versus contact sports (basketball, soccer, lacrosse, etc.) is important because of the relative risk of repetitive and traumatic hyperextension and axial compression forces to the cervical spine. In our opinion, collision sports such as American football represent a significantly higher risk of re-injury in players with stenosis than contact sports.

Based on the most recent literature, we believe that the measurement of the cervical spinal canal diameter on sagittal T2 MRI provides the most specific information in the diagnosis of this condition. Although historically cervical spinal stenosis has been defined as a space available for the cord < 14 mm on plain radiographs,[41] more recent and pertinent data suggest that this threshold may be too conservative when it comes to the risk of a spinal cord injury in the American football population. The data from Schroeder et al suggests that 10-mm sagittal canal diameter is compatible with successful NFL careers without increased risk of neurologic sequelae despite a small sample size.[42] Furthermore, Aebli et al demonstrated that sagittal canal diameter of < 8 mm on MRI increases the risk of acute spinal cord injury after minor trauma to the cervical spine.[15] Although there are certainly other data that utilizes other measures of spinal canal space,[43] most of the evidence-based literature points towards this range as a potential threshold for return-to-play criteria when treating these athletes. In a systematic review of clinical studies involving high-energy contact athletes, Dailey et al strongly recommended that those without stenosis can return to play and weakly recommend that those with stenosis not return to play after one episode of transient cervical cord neurapraxia.[3]

In players with cervical stenosis, absolute contraindications of RTP[4] include persistent neurological symptoms, cervical pain, loss of ROM, os odontoideum, basilar invagination, atlanto-occipital instability, Chiari malformation, functional spinal stenosis, cervical instability due to fracture or ligamentous disruption, symptoms lasting >36 hours, or repeat episodes of transient quadriplegia [4,34,44]. RTP recommendations in NFL athletes who demonstrate persistent cord hyperintensity despite surgical treatment is controversial.[45] While the presence of residual edema in the spinal cord may represent smaller reserve to withstand future forces on the cervical spine, Maroon et al recommends that cord hyperintensity after surgery may not preclude RTP in NFL athletes.[12,20] Tempel et al also reported persistent T2 hyperintensity in 3 professional athletes, all of whom returned to play without re-injury.[12] In their modified Delphi analysis, Schroeder et al found strong consensus among physician regarding RTP in asymptomatic patients with no signal changes and canal diameter >10mm or resolved signal changes and canal diameter >13mm. Those with no history of signal changes and canal diameters <10mm should be considered on case-by-case basis and those with resolved signal changes and canal diameters <10mm should not RTP. No consensus was reached for patients with persistent signal changes.[33]

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In summary, RTP indications for patients with a history of congenital cervical stenosis or transient quadriplegia are controversial. Imaging studies should be considered in context with the athlete's history, mechanism of injury, physical examination, and discussion of specific RTP risks with the athletes.[13,46,47]

Cervical Disc Herniation

CDH, which is a career-threatening injury, is more common in NFL and other contact athletes relative to the general population.[1,48] The injury is associated with upper extremity radiculopathy, pain with range of motion, and/or coordination difficulties. Players without significant nerve root compression may be treated conservatively and return to play upon regaining full cervical motion and upper extremity strength.[9,49]

Both non-operative and operative treatments of CDH have been proven to be successful, which indicates that players with this condition in the absence of neurologic deficits or myelopathy can heal without aggressive treatment. For example, in their review of 16 NFL athletes with CDH, Meredith et al describes 8 players who were treated non-operatively with therapy, NSAIDs, or epidural steroid injections and successfully returned to play.[49] Hsu et al reviewed 99 NFL athletes with CDH and found that although operative management was associated with higher RTP compared to non-operative management, performance scores and numbers of games started was not different between treatment groups.[18] As with cases of CDH in non-football athletes, initial management of CDH should include a trial of non-operative management which could include epidural injections and anti-neuropathic agents.

For patients with persistent pain, significant extremity weakness, or cervical stenosis are present, operative management has demonstrated successful RTP. While anterior cervical discectomy and fusion is the most common operation offered to these players, posterior foraminotomies can also be successful for appropriate pathology. Recent data suggest that players who undergo a foraminotomy need to be counseled as to the significant increased risk of reoperation compared to a fusion, but depending on the goals of the player and medical team, this can be a reasonable option. An athlete who has undergone a single-level anterior or posterior fusion who is asymptomatic, neurologically intact, without pain or ROM restriction, and evidence of radiographic fusion, can be cleared to return to sport;[49] however, the athlete may be at increased risk of repeated herniation above or below the fused level.[50] This recommendation is supported by Schroeder et al in their Delphi analysis. However, when considering asymptomatic players without symptoms, no T2-signal changes, and solid 2-level ACDF fusions, the authors found no consensus among team physicians regarding RTP.[33] Similarly, they found only weak consensus among experts regarding the case-by-case consideration needed for athletes undergoing corpectomy or posterior cervical surgery.[33]

Relative contraindications to RTP include athletes with an asymptomatic pseudarthrosis, 2- and 3-level fusion, and/or persistent signal intensity of T2 imaging.[9] Absolute contraindications to RTP include persistent neurologic findings, ROM limitations, pain, 4+ level anterior or posterior fusions.[9]

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Cervical Fractures

Cervical fractures can occur when the area between the vertebral body and decelerating head undergoes rapid compression, requiring as little as 150ft lbs of force to fracture the cervical vertebrae.[29,51,52] Certain fracture patterns involving the spinous process or unilateral lamina do not typically cause significant instability and require only hard collar immobilization for a period of 6-12 weeks as definitive treatment. [44] Stability on flexion-extension radiographs after healing and maintenance of cervical lordosis are required for RTP.[29,51] Unstable fractures are best treated with surgical fixation,[27,29,47] and in general, the literature supports return to play for players who are treated with a one-level subaxial anterior or posterior fusion if there is no pain, full ROM, normal muscle strength, complete fusion on imaging, and a normal neurological exam.[13]

The decision to allow an athlete to RTP is highly dependent upon surgeon and pattern of injury. For example, in their case-based survey study, France et al found that 12% of spine surgeons would not allow a collegiate football player with a non-operatively healed C5 burst fracture to return to football.[39] In their case-based questionnaire of 113 sports and spine physicians, Morganti et al. found similar disagreement in management of certain subaxial cervical spine injuries treated with one-level fusion.[8] Schroeder et al found that unanimous consensus among an expert panel of spine surgeons regarding RTP in athletes with solid fusion after burst, compression, or facet fracture with no instability, no T2 signal change, and solid fusion.[33]

Despite some disagreements, there appears to be strong consensus regarding certain absolute contraindications, including any patient with a fracture with residual neurological deficits, underlying congenital stenosis, cervical laminectomy, and any procedures that require occipital, C1-C2, C2-C3, or 3+ level fusions.[29,53,54] Consideration for NFL athlete RTP should involve imaging demonstrating solid fusion in the area of injury, lack of instability on flexion and extension radiographs, maintenance of normal cervical lordosis, full ROM with no associated pain, and preinjury muscle strength. [29,44,53] Additionally, sagittal misalignment >11° when compared to other levels for isolated burst fractures and fractures that disrupt the posterior bony or ligamentous tension band may preclude RTP. [44,53]

Our ability to develop generalizable recommendations for RTP in American football athletes after cervical spine injury is with substantial limitation. The majority of primary sources used for developing RTP recommendations were cases studies, panel discussions, or review of general injury data that is limited by lack of clinical granularity. Although certain recommendations are strongly supported by the literature (athletes with persistent neurological symptoms should not RTP), others are less robust or even controversial. In these cases, individual surgeons must rely upon experience and patient-specific factors for making RTP decisions. Given the lack of high-level evidence directing RTP after cervical spine injury, there is a theoretical risk that this consensus-based approach to managing cervical spine injuries may become more dogma than guideline. However, we acknowledge that the purpose of this review is to generate guiding principles rather than rules for the management of cervical spine injuries in athletes and to encourage future reseach efforts aimed at improving the evidence for RTP.

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The development of strong recommendations for RTP and cervical spine injuries in football athletes requires a strong study design, prospective collection, and outcomes data both on and off the field. Major professional sport organizations all have this mechanism in place, however, the approach and type of allowable data to be collected varies widely[55]. We believe that if medical staff of professional sports are allowed to prospectively collect data on spine injuries in athletes and assess post-treatment outcomes through patient-reported outcomes and statistical performance then the controversial questions in this population that spine physicians struggle with can be answered. While some major North American sport leagues currently allow for this, many critical ones do not.

Conclusion

Although an improved understanding of injury mechanisms as well as the evolution of rule changes have decreased its incidence, cervical spine injuries continue to occur frequently in American football. These conditions present a myriad of challenges and can result in significant consequences for players with improper management. RTP guidelines after injury need to be assessed based on a case-by-case basis, but we believe in several core principles. Consideration for RTP should be individualized based upon a player's history, physical examination, imaging characteristics, and sport/position played. Players should have full ROM of the cervical spine without associated pain, return of normal muscle strength, and an intact neurologic exam. Recommendations regarding congenital issues in the cervical spine and postsurgical criteria for RTP continue to evolve with the continued research of cervical spine injuries in NFL athletes.

KEY POINTS

- Elite American football athletes are a unique patient population who are at increased risk for cervical spine injury.
- This study proposes a set of evidence-based guidelines for return to play after cervical spine injury in American football athletes.
- These guidelines incorporate general guiding principles, best available evidence, and consensus opinion among experts.
- This review and consensus work will guide spine surgeons and sports practitioners in the management of American football athletes with cervical spine injuries.

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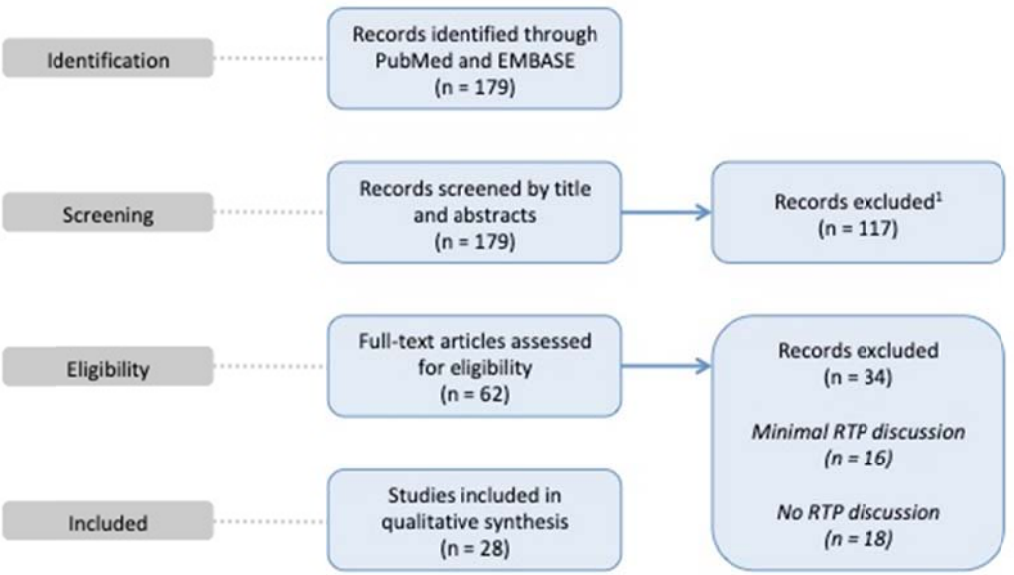
Figure 1: Study Selection Process

Source: PubMed, EMBASE, published or electronically available prior to December 1, 2019.

Note: 1. Reasons for exclusion include no full text available (n=13), unrelated topics (n=32), high school football players only or rugby players only (n=20), letter or opinion articles (n=4), or literature reviews (n=45). Abbreviations: RTP, return to play.



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